

[9/A9]

Roll No. _____

No. of Questions: 2

SARDAR PATEL UNIVERSITY

External Examination

F.Y.BCA (Sem-1)

US01EBCA01: Digital Computer Electronics

Time: 10.00 am – 12.00 noon

Marks: 70

Date: 08-11-2017

Wednesday

Q-1 Select the correct option from the following questions.

10

- 1 An invert gate is also called a _____ gate.
A. NOR B. NOT C. XNOR D. NAND
- 2 De Morgan's first theorem says that a NOR gate is equivalent to a _____.
A. bubbled OR B. bubbled NOR C. bubbled AND D. AND bubbled
- 3 In Comparator, _____ gate is use for comparing bits in word.
A XOR B AND C NOR D XNOR
- 4 In k-map, pair eliminates _____ variable.
A. one B. two C. three D. four
- 5 A _____ is a memory element that stores a binary digit.
A. binary adder B. decoder C. D flipflop D. multiplexer
- 6 A gate is a logic circuit with one or more input signals but only _____ output signal.
A. two B. one C. three D. four
- 7 A register is a group of _____ that work together as a unit.
A. flip-flop B. decoder C. multiplexer D. gates
- 8 Ring counter producing words with 1 high bit, which shifts _____ position per clock pulse
A. one B. two C. three D. none
- 9 _____ is way to simplify the equation.
A. Boolean Algebra B. K-MAP C. BOTH D. NONE
- 10 In half adder XOR gate's output is.....
A. carry B. sum C. reminder D. none

Q-2 Answer the following questions. (ATTEMPT ANY TEN)

20

- 1 Write truth table for : $A'B+B'C$
- 2 Describe the NAND, NOR gates.
- 3 Explain distributive law.
- 4 Describe pair in k-map.

(1)

(P.T.O)

- 5 Define encoder and decoder.
- 6 Explain sum of product.
- 7 Draw the circuit diagram of Half Adder.
- 8 What are the differences between Half and Full Adder?
- 9 Define Multiplexer. What is the use it?
- 10 Define : register and shift register. What are the types of shift register?
- 11 Draw timing diagram of Ring Counter.
- 12 Draw the circuit diagram of Controlled Buffer Register.

- Q-3 A Explain associative and commutative law. 6
- B Explain AND, OR gates with diagram and truth table. 4

OR

- Q-3 A Write note on: De'Morgan's first and second theorem. 6
- B Explain XOR and NOT gates with diagram and truth table. 4

- Q-4 A Explain 8x3 line encoder in detail. 5
- B Explain comparator with example. 5

OR

- Q-4 A Explain 3x8 line decoder in detail. 6
- B What is k-map? Explain octet with example. 4

- Q-5 A Explain Half adder in detail. 4
- B Explain 4x1 multiplexer. 6

OR

- Q-5 A Explain Full adder in detail. 4
- B Explain 4 bit binary subtractor with circuit diagram. 6

- Q-6 Explain shift left and shift right registers in detail. 10

OR

- Q-6 Explain D flip-flop (clocked and unclocked) in detail. 10

Best of luck

(2)